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Research Inspirations Behind CedarDB: Insights from LeanStore, Umbra, and Colibri

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Abstract

This essay explores the evolution of high-performance storage engines culminating in CedarDB, a commercial, state-of-the-art database system built on over a decade of academic research. Specifically, it examines three foundational papers that contribute to the technological lineage of CedarDB: LeanStore, a lightweight and scalable buffer manager optimized for modern hardware; Umbra, a disk-based database system that extends LeanStore's principles with variable-size pages and full system integration to achieve in-memory performance; and Colibri, a hybrid cloud storage engine optimized for HTAP (Hybrid Transactional and Analytical Processing) workloads. These works collectively inspire CedarDB's design to make full use of modern multi-core CPUs, RAM, and SSDs. By tracing these innovations and their academic origins, the essay analyzes how CedarDB achieves scalable performance for data-intensive workloads in hybrid memory and storage environments. It also discusses the broader social and environmental implications of such systems, particularly their roles in promoting efficiency, sustainability, and accessibility in cloud-native deployments. Consequently, CedarDB represents a pivotal step toward practical, scalable, and sustainable database systems for modern hybrid workloads.

Keywords: CedarDB, Colibri, Umbra, LeanStore, Database, DBMS

1. Introduction

CedarDB (2) builds directly upon three key research systems: LeanStore, which optimized buffer management for in-memory and beyond-main-memory performance; Umbra, which extended these ideas into a full disk-based engine; and Colibri, which added cloud-native, hybrid OLTP/OLAP capabilities. Together, these systems provide the architectural foundation for CedarDB's performance, scalability, and HTAP functionality.

2. Technical Foundations of CedarDB.

2.1. LeanStore: A Lightweight Foundation for Buffer Management in CedarDB.

LeanStore (4) introduced a scalable, pointer-based buffer management system that minimized the overhead of managing pages in memory or on disk. By using pointer swizzling to replace logical page IDs with direct memory pointers, and a cooling mechanism for eviction, LeanStore was able to deliver in-memory-like performance even as datasets exceeded RAM.

These innovations directly influenced CedarDB's memory management layer, particularly its approach to reducing synchronization and eviction overhead across multiple threads. CedarDB extends LeanStore's ideas by adapting them to work with variable-sized pages and hybrid memory environments.

Problem

Traditional buffer managers add significant overhead even when

data fits in RAM, due to costly page table lookups, locking and eviction tracking. In-memory systems avoid these costs but fail when datasets grow beyond RAM.

Contributions

Pointer swizzling replaces page IDs with direct memory pointers when pages are resident in RAM, allowing for faster data access. Lightweight eviction, or "cooling," randomly selects pages to be marked as cooling and reactivates them if accessed again, which reduces the overhead of precise tracking. Finally, scalable synchronization employs optimistic latching and epoch-based reclamation to minimize contention among threads.

Results

LeanStore achieves the performance nearly equivalent to pure in-memory systems when the working set fits in RAM. Its performance degrades gracefully when spilling to SSD, outperforming conventional buffer managers like WiredTiger (7) or BerkeleyDB (1). The system also demonstrates efficient scalability across multi-core CPUs.

Evaluation

LeanStore proves that a buffer manager can be designed to support both in-memory speed and larger-than-memory datasets without the traditional cost overhead.

2.2. Umbra: Disk-Based Performance at In-Memory Speed.

Umbra (5) integrated LeanStore's buffer management into a complete disk-based database system. Its key innovations included the use of virtual memory management with OS-level mechanisms

such as *mmap* and *madvise* to handle large address spaces, as well as the introduction of variable-size pages and versioned latches to improve concurrency.

CedarDB’s core architecture reuses these ideas, particularly the unified buffer pool and adaptive execution engine. While Umbra was primarily academic, CedarDB applies these techniques in production with broader workload support and deployment options.

Problem

Even with LeanStore, supporting complex database features (e.g., variable-size objects, strings, adaptive queries) at in-memory performance requires deeper architectural changes.

Contributions

The system employs a unified buffer pool with variable-size pages, allowing different page sizes to improve space utilization and performance. It uses virtual memory management through *mmap* and other OS-level features to handle page frames, reduce fragmentation, and manage eviction efficiently. Versioned latches enable concurrent page access with minimal synchronization overhead. Additionally, the system includes optimized subsystems for string handling, temporary memory management, adaptive query execution, and profiling.

Results

Umbra delivers performance comparable to HyPer (3), a pure in-memory system, on many workloads when the data fits entirely in RAM. The system effectively handles SSD-backed cold data, maintaining high throughput and stable performance. Even under heavy load, the system maintains consistent speed due to its efficient eviction mechanisms and optimized I/O handling.

Evaluation

Umbra combines storage and execution optimizations to create a general-purpose database that bridges the gap between in-memory and disk-based systems, offering low overhead and high throughput.

2.3. Colibri: HTAP in the Cloud with Hot/Cold Separation.

Colibri (6), built on top of the Umbra storage layer, was designed to support Hybrid Transactional and Analytical Processing (HTAP). Its main contributions included a hybrid row-column storage layout that separates hot (OLTP) and cold (OLAP) data, support for cloud object storage with asynchronous I/O, and a reduction in logging overhead for cold or archived data.

CedarDB integrates Colibri’s hybrid storage layout and cloud-readiness features, positioning it as a cloud-native HTAP database. This makes it suitable for real-time analytics on transactional data, even when datasets exceed memory and are distributed across cloud storage.

Problem

Modern data-intensive applications (e.g., SaaS, finance, real-time dashboards) need to analyze transactional data without delay. Classical OLTP and OLAP separation increases complexity and latency. Additionally, cloud-native storage presents new challenges like high I/O latency, limited RAM, and multi-tiered

storage.

Contributions

The system employs a hybrid row-column storage model that combines row-based hot data for fast writes and updates with columnar cold data optimized for analytical scans. Through hot and cold tiering, it identifies data access patterns and migrates infrequently used data to slower, more cost-effective cloud storage such as object stores. Asynchronous I/O and data compression techniques allow the system to overlap computation with input/output operations while conserving bandwidth and storage space. Finally, logging and consistency optimizations reduce overhead without compromising transactional correctness or analytical integrity.

Results

Colibri achieves up to a tenfold performance improvement on HTAP workloads compared to traditional database engines. The system provides efficient access to cold data stored on cloud object stores, such as Amazon S3, while maintaining minimal impact on the performance of hot OLTP operations. Furthermore, the system scales effectively as dataset sizes and workload concurrency increase, ensuring consistent and reliable performance under growing demands.

Evaluation

Colibri shows how a carefully designed storage engine can support cloud-native HTAP by separating responsibilities and optimizing both transactional and analytical access patterns.

2.4. CedarDB: Integration of Research into a Production System

CedarDB draws upon key ideas from prior research systems, including LeanStore’s efficient buffer management concepts, Umbra’s flexible page architecture and adaptive execution, and Colibri’s hybrid storage and cloud-oriented design. Rather than directly incorporating these systems, CedarDB reimplements and extends their principles within a new, production-grade architecture. It combines these research-driven techniques with enterprise-level features such as high availability, security, and user-oriented interfaces.

As a result, CedarDB functions as a general-purpose, high-performance HTAP database that efficiently scales from small transactional workloads to large analytical platforms while maintaining robustness suitable for commercial deployment.

3. Social and Environmental Aspects.

The social and environmental implications of systems like CedarDB extend beyond their technical achievements. On one hand, their efficiency and ability to process massive data volumes with less hardware translate into reduced operational energy and lower infrastructure costs. By enabling real-time analytics, such systems also empower organizations to make faster, data-driven decisions across sectors such as healthcare, logistics, and public policy. On the other hand, the increasing centralization of data infrastructure raises concerns about energy consumption in large-scale cloud environments, electronic performance, and automation can unintentionally exacerbate inequalities, through

data privacy risks or the environmental burden of continuous hardware upgrades. Achieving genuine sustainability therefore requires balancing performance ambitions with ethical responsibility, transparent governance, and lifecycle awareness in system design.

3.1. Social Impacts

Among the positive impacts, businesses and organizations can make decisions faster through real-time analytics. Another benefit is the reduction of infrastructure needs by avoiding the over-provisioning of RAM. These capabilities also support innovation by enabling new applications in finance, healthcare, logistics, and other fields that require simultaneous access to real-time and historical data.

At the same time, real-time access to large volumes of data raises privacy concerns, as it increases the risks of surveillance and data misuse. More sophisticated processing engines can also heighten system complexity, making maintenance and auditing more difficult and potentially affecting reliability or fairness. Furthermore, strong dependence on cloud-based systems (such as those used in Colibri) can lead to centralization, vendor lock-in, and reduced user control over data.

Beyond social implications, the environmental footprint of such systems also warrants careful consideration.

3.2. Environmental Considerations

From an environmental perspective, efficient resource use helps minimize waste by dynamically managing which data must be kept in memory, thereby saving energy and reducing costs. Storing cold data in compressed formats further reduces storage requirements and I/O overhead.

However, increasing reliance on SSDs and DRAM can contribute to higher levels of electronic waste and greater energy consumption. The use of cloud infrastructure, such as object stores like S3, also carries environmental costs due to the power and cooling demands of data centers. Moreover, always-on analytics in HTAP workloads can further increase server workloads and energy usage, particularly when systems are not properly optimized.

4. Conclusion

The progression from LeanStore to Umbra and Colibri reflects a shift from traditional assumptions about memory and storage toward more adaptive, tiered, and cloud-aware database designs. LeanStore introduced efficient buffer management techniques for modern hardware, while Umbra built upon these ideas by introducing variable-size pages and integrating them into a full-featured database system. Colibri then extended these principles to the cloud era with hybrid transactional and analytical processing. Together, these research systems illustrate how modern databases can remain both efficient and scalable as data volumes exceed main memory and workloads increasingly demand real-time insights.

However, higher performance also introduces trade-offs, including greater architectural complexity, environmental costs, and challenges in cloud dependency and data governance. Future research may focus on automated data tiering, energy-aware execution, and privacy-preserving HTAP systems to better balance speed, cost, and responsibility.

CedarDB builds upon this broader research lineage, not as a direct continuation, but as a commercial system that re-imagines and integrates these innovations. It stands as both a consolidation of a decade of academic progress and a step toward a future where efficiency, sustainability, and responsibility converge.

References

- [1] BERKELEYDB. Oracle berkeley db. <https://www.oracle.com/se/database/technologies/related/berkeleydb.html>. Accessed: 2025-09-10.
- [2] CEDARDB. Cedardb: One database, endless possibilities. <https://cedardb.com/>. Accessed: 2025-09-10.
- [3] HYPER. Hyper – a hybrid oltpolap high performance dbms. <https://hyper-db.de/>. Accessed: 2025-09-10.
- [4] LEIS, V., HAUBENSCHILD, M., KEMPER, A., AND NEUMANN, T. Leanstore: In-memory data management beyond main memory. In *2018 IEEE 34th International Conference on Data Engineering (ICDE)* (2018), IEEE, pp. 185–196.
- [5] NEUMANN, T., AND FREITAG, M. J. Umbra: A disk-based system with in-memory performance. In *CIDR* (2020), vol. 20, p. 29.
- [6] SCHMIDT, T., DURNER, D., LEIS, V., AND NEUMANN, T. Two birds with one stone: Designing a hybrid cloud storage engine for htap. *Proceedings of the VLDB Endowment* 17, 11 (2024), 3290–3303.
- [7] WIREDTIGER. Wiredtiger storage engine. <https://www.mongodb.com/docs/manual/core/wiredtiger/>. Accessed: 2025-09-10.